The 4 Goals of AI

The big idea

AI can be built with 4 different goals. Just ask two simple questions:

1. Do I want the machine to **think** or to **act**?
2. Do I want it to be **like a human** or to be **perfectly correct (rational)**?

That gives 4 combinations.

The 4 goals, with easy way of understanding

1. Thinking humanly: "Think like a human"

The machine's brain should work the way a human brain works. It doesn't matter only whether the answer is right, but how it got there.

Example: You solve 12 × 15 by breaking it into 10 × 15 + 2 × 15. A machine copying this same step-by-step method is "thinking humanly."

2. Thinking rationally: "Think correctly"

The machine follows strict logic, with no feelings and no guessing.

Example: All birds have wings. A sparrow is a bird. So a sparrow has wings. The machine always reasons like this.

3. Acting humanly: "Behave like a human"

It doesn't matter what's going on inside. If you talk to the machine and can't tell whether it's a human or a machine, it passes. This is the Turing Test.

Example: A chatbot that chats so naturally you think a real person is typing.

4. Acting rationally: "Do the best thing"

The machine looks at the situation and does whatever gives the best result.

Example: A self-driving car choosing the safest and fastest route. It doesn't copy a human driver, it just does what works best.

Difference between acting and thinking

	Like a human	**Perfectly correct**
Thinking	Copy how humans think	Use pure logic
Acting	Copy how humans behave	Do the best action

The differences in one line each

- **Thinking vs. acting:** Thinking is about what happens inside the brain. Acting is about what you see from outside.
- **Humanly vs. rationally:** Humanly means copying humans, mistakes included. Rationally means doing the ideal, perfect thing.

Most AI used is

Acting rationally. It's the easiest to measure and the most useful, because we don't need to copy human flaws, we just want the machine to get the best result.